Total-Evidence Dating under the Fossilized Birth-Death Process

Chi Zhang[1], Tanja Stadler[2, 3], Seraina Klopfstein[1, 4], Tracy A. Heath[5, 6, 7], and Fredrik Ronquist[1, *]

[1]Department of Bioinformatics and Genetics, Swedish Museum of Natural History, SE-10405 Stockholm, Sweden; [2]Department of Environmental Systems Science, Eidgenössische Technische Hochschule Zürich, 4053 Basel, Switzerland; [3]Swiss Institute of Bioinformatics (SIB), Switzerland; [4]Department of Invertebrates, Natural History Museum Bern, CH-3005 Bern, Switzerland; [5]Department of Integrative Biology, University of California, Berkeley, CA 94720 USA; [6]Department of Ecology and Evolutionary Biology, University of Kansas, Lawrence, KS 66045 USA; [7]Department of Ecology, Evolution & Organismal Biology, Iowa State University, Ames, IA 50011 USA;

∗ Correspondence to be sent to: Department of Bioinformatics and Genetics, Swedish Museum of Natural History, Box 50007, SE-10405 Stockholm, Sweden;
E-mail: fredrik.ronquist@nrm.se

Supplemental Tables

Table S1: Probability of fossils being ancestral under the IGR model and different versions of the FBD model. For each fossil, we show the name, the completeness of morphological coding, and the probability of being ancestral (mean and standard deviation across four independent runs) under piecewise-constant FBD and random sampling (pcFBD_rnd), piecewise-constant FBD and diversified sampling (pcFBD_div), constant-rate FBD and random sampling (FBD_rnd), and constant-rate FBD and diversified sampling (FBD_div), respectively.

		pcFBD_rnd		pcFBD_div		FBD_rnd		FBD_div	
Name	%	Mean	SD	Mean	SD	Mean	SD	Mean	SD
Triassoxyela	14	0.582	0.014	0.016	0.001	0.006	0.000	0.010	0.000
Asioxyela	6	0.562	0.005	0.016	0.001	0.005	0.001	0.010	0.001
Nigrimonticola	12	0.015	0.001	0.000	0.000	0.000	0.000	0.000	0.000
Gigantoxyelinae	20	0.106	0.027	0.001	0.000	0.000	0.000	0.001	0.001
Spathoxyela	20	0.372	0.015	0.005	0.001	0.001	0.000	0.005	0.001
Xyela_mesozoica	12	0.817	0.007	0.040	0.001	0.009	0.001	0.038	0.006
Angaridyela	20	0.005	0.001	0.000	0.000	0.000	0.000	0.000	0.000
Xyelotoma	10	0.004	0.002	0.001	0.000	0.000	0.000	0.000	0.000
Undatoma	19	0.217	0.023	0.006	0.001	0.000	0.000	0.004	0.001
Dahuratoma	16	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Mesolyda	14	0.020	0.006	0.001	0.000	0.000	0.000	0.000	0.000
Turgidontes	10	0.586	0.024	0.013	0.001	0.002	0.000	0.013	0.001
Aulidontes	10	0.336	0.010	0.013	0.002	0.001	0.000	0.010	0.001
Protosirex	17	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Aulisca	10	0.428	0.013	0.013	0.001	0.001	0.000	0.011	0.001
Anaxyela	14	0.236	0.023	0.008	0.001	0.001	0.000	0.008	0.001
Syntexyela	14	0.254	0.017	0.017	0.001	0.003	0.000	0.018	0.001
Karatavites	14	0.003	0.001	0.000	0.000	0.000	0.000	0.000	0.000
Stephanogaster	12	0.096	0.013	0.002	0.001	0.000	0.000	0.002	0.001
Leptephialtites	11	0.472	0.012	0.019	0.002	0.002	0.001	0.016	0.002
Cleistogaster	19	0.453	0.036	0.061	0.011	0.001	0.000	0.048	0.013
Sepulca	10	0.436	0.027	0.018	0.001	0.002	0.000	0.017	0.001
Onochoius	10	0.071	0.006	0.003	0.000	0.000	0.000	0.003	0.000
Ghilarella	12	0.004	0.001	0.000	0.000	0.000	0.000	0.000	0.000
Paroryssus	10	0.699	0.030	0.125	0.006	0.004	0.001	0.108	0.018
Praeoryssus	10	0.034	0.003	0.003	0.000	0.000	0.000	0.003	0.002
Mesorussus	4	0.250	0.009	0.006	0.003	0.000	0.000	0.005	0.000
Trematothorax	15	0.558	0.039	0.034	0.003	0.003	0.001	0.032	0.004
Thoracotrema	11	0.012	0.003	0.001	0.000	0.000	0.000	0.001	0.000
Prosyntexis	12	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Kulbastavia	14	0.395	0.015	0.013	0.002	0.003	0.000	0.012	0.001
Brachysyntexis	12	0.443	0.010	0.026	0.002	0.005	0.001	0.026	0.002
Symphytopterus	16	0.094	0.011	0.004	0.001	0.000	0.000	0.003	0.001
Eoxyela	14	0.817	0.004	0.082	0.004	0.016	0.001	0.077	0.005
Liadoxyela	9	0.283	0.022	0.007	0.001	0.001	0.000	0.006	0.001
Abrotoxyela	14	0.652	0.003	0.016	0.002	0.005	0.001	0.016	0.001
Pseudoxyelocerus	5	0.807	0.014	0.092	0.007	0.009	0.001	0.091	0.008
Palaeathalia	7	0.248	0.020	0.005	0.000	0.001	0.000	0.005	0.001
Ferganolyda	5	0.784	0.008	0.047	0.005	0.011	0.002	0.049	0.002

PamphiliidaeUndesc	7	0.679	0.011	0.092	0.012	0.005	0.001	0.063	0.014
Rudisiricius	14	0.780	0.009	0.081	0.007	0.008	0.001	0.079	0.033
Sogutia	4	0.656	0.017	0.037	0.002	0.006	0.001	0.038	0.004
Xyelula	11	0.040	0.005	0.003	0.002	0.000	0.000	0.002	0.001
Brigittepterus	5	0.122	0.010	0.003	0.001	0.000	0.000	0.003	0.000
Grimmaratavites	10	0.629	0.022	0.162	0.006	0.004	0.000	0.121	0.006

Table S2: Probability of fossils being ancestral under the TK02 model. See also Table S1.

		pcFBD_rnd		pcFBD_div		FBD_rnd		FBD_div	
Name	%	Mean	SD	Mean	SD	Mean	SD	Mean	SD
Triassoxyela	14	0.051	0.003	0.001	0.000	0.001	0.000	0.000	0.000
Asioxyela	6	0.045	0.003	0.001	0.000	0.001	0.000	0.000	0.000
Nigrimonticola	12	0.001	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Gigantoxyelinae	20	0.019	0.002	0.000	0.000	0.000	0.000	0.000	0.000
Spathoxyela	20	0.006	0.001	0.000	0.000	0.000	0.000	0.000	0.000
Xyela_mesozoica	12	0.044	0.004	0.000	0.000	0.000	0.000	0.000	0.000
Angaridyela	20	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Xyelotoma	10	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Undatoma	19	0.004	0.001	0.000	0.000	0.000	0.000	0.000	0.000
Dahuratoma	16	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Mesolyda	14	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Turgidontes	10	0.031	0.002	0.000	0.000	0.000	0.000	0.000	0.000
Aulidontes	10	0.015	0.002	0.000	0.000	0.000	0.000	0.000	0.000
Protosirex	17	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Aulisca	10	0.032	0.003	0.000	0.000	0.000	0.000	0.000	0.000
Anaxyela	14	0.009	0.002	0.000	0.000	0.000	0.000	0.000	0.000
Syntexyela	14	0.021	0.001	0.000	0.000	0.000	0.000	0.000	0.000
Karatavites	14	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Stephanogaster	12	0.004	0.001	0.000	0.000	0.000	0.000	0.000	0.000
Leptephialtites	11	0.015	0.002	0.000	0.000	0.000	0.000	0.000	0.000
Cleistogaster	19	0.018	0.002	0.000	0.000	0.000	0.000	0.000	0.000
Sepulca	10	0.035	0.002	0.000	0.000	0.000	0.000	0.000	0.000
Onochoius	10	0.007	0.001	0.000	0.000	0.000	0.000	0.000	0.000
Ghilarella	12	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Paroryssus	10	0.052	0.003	0.001	0.000	0.001	0.000	0.000	0.000
Praeoryssus	10	0.001	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Mesorussus	4	0.014	0.001	0.000	0.000	0.000	0.000	0.000	0.000
Trematothorax	15	0.043	0.002	0.000	0.000	0.001	0.000	0.000	0.000
Thoracotrema	11	0.001	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Prosyntexis	12	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Kulbastavia	14	0.037	0.004	0.000	0.000	0.000	0.000	0.000	0.000
Brachysyntexis	12	0.052	0.004	0.000	0.000	0.001	0.000	0.000	0.000
Symphytopterus	16	0.003	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Eoxyela	14	0.055	0.003	0.000	0.000	0.001	0.000	0.000	0.000
Liadoxyela	9	0.011	0.001	0.000	0.000	0.000	0.000	0.000	0.000
Abrotoxyela	14	0.039	0.002	0.000	0.000	0.000	0.000	0.000	0.000
Pseudoxyelocerus	5	0.052	0.003	0.001	0.000	0.001	0.000	0.000	0.000
Palaeathalia	7	0.011	0.001	0.000	0.000	0.000	0.000	0.000	0.000
Ferganolyda	5	0.056	0.004	0.001	0.000	0.001	0.000	0.000	0.000
PamphiliidaeUndesc	7	0.042	0.002	0.001	0.000	0.001	0.000	0.000	0.000
Rudisiricius	14	0.053	0.003	0.001	0.000	0.001	0.000	0.000	0.000
Sogutia	4	0.046	0.003	0.001	0.000	0.001	0.000	0.000	0.000
Xyelula	11	0.002	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Brigittepterus	5	0.004	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Grimmaratavites	10	0.027	0.004	0.000	0.000	0.000	0.000	0.000	0.000

Table S3: Comparing parameter estimates (mean and 95% HPD) from MrBayes and BEAST2. To make the inference models exactly the same (but not necessarily realistic), we used HKY+Γ_4 for the molecular partitions and unordered states without rate variation among characters for the morphology partition. The fossil ages were fixed to the midpoint of the corresponding stratum. We used strict clock model with lognormal (-7, 0.5) for the clock rate and constant-rate FBD prior with random sampling of extant taxa (Diversified sampling is currently unavailable in BEAST2). The other settings followed the descriptions in section "Total-evidence dating of Hymenoptera". The minimal ESS is >300.

Parameters	MrBayes	BEAST2
Root Age	771.2 (590.3, 979.4)	765.4 (585.6, 961.7)
Age of Hymenoptera	614.3 (473.3, 777.6)	613.6 (474.7, 765.9)
Clock Rate	4.39E-4 (3.31E-4, 5.52E-4)	4.36E-4 (3.30E-4, 5.46E-4)
Diversification (*d*)	0.0090 (0.0070, 0.011)	0.0091 (0.0070, 0.011)
Turnover (*r*)	0.986 (0.981, 0.991)	0.986 (0.981, 0.991)
FossilSamplingProp (*s*)	1.08E-5 (1.0E-5, 1.24E-5)	1.08E-5 (1.0E-5, 1.25E-5)
Prop_AncestralFossils	0.0038 (0.0, 0.022)	0.0038 (0.0, 0.022)
*m*_morphology	0.89 (0.83, 0.96)	1.01 (0.93, 1.09)
*m*_12S16S	1.60 (1.41, 1.81)	1.30 (0.95, 1.60)
*m*_18S	0.14 (0.12, 0.16)	0.14 (0.12, 0.17)
*m*_28S	0.28 (0.22, 0.33)	0.29 (0.23, 0.34)
*m*_CO1_12	0.79 (0.72, 0.86)	0.85 (0.77, 0.94)
*m*_EF1a_12	0.24 (0.21, 0.27)	0.26 (0.22, 0.29)
*m*_ EF1a_3	4.03 (3.85, 4.21)	4.03 (3.82, 4.25)
κ_12S16S	4.67 (4.02, 5.47)	4.15 (3.30, 5.07)
κ_18S	4.69 (3.92, 5.51)	4.70 (3.89, 5.48)
κ_28S	3.72 (3.07, 4.44)	3.73 (3.04, 4.45)
κ_CO1_12	2.81 (2.57, 3.07)	2.82 (2.57, 3.08)
κ_EF1a_12	2.59 (2.24, 2.96)	2.61 (2.27, 3.00)
κ_ EF1a_3	6.44 (5.97, 6.95)	6.28 (5.81, 6.76)
α_12S16S	0.44 (0.38, 0.49)	0.46 (0.40, 0.54)
α_18S	0.21 (0.18, 0.24)	0.19 (0.16, 0.21)
α_28S	0.23 (0.20, 0.26)	0.20 (0.18, 0.23)
α_CO1_12	0.29 (0.26, 0.32)	0.27 (0.24, 0.30)
α_EF1a_12	0.15 (0.13, 0.16)	0.13 (0.12, 0.14)
α_ EF1a_3	1.05 (0.93, 1.17)	1.01 (0.91, 1.12)

Supplemental Figures

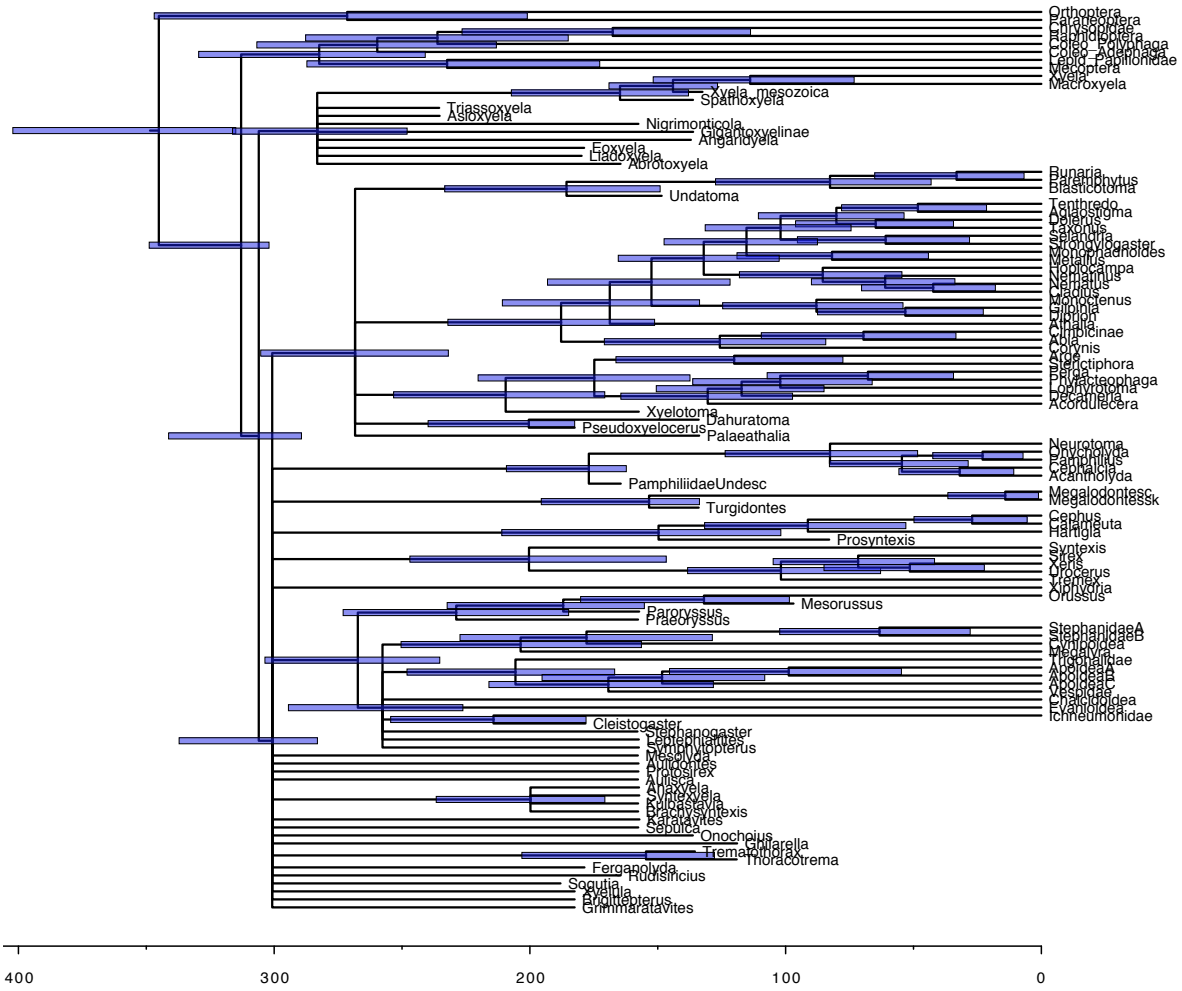


Figure S1: Majority-rule consensus tree from total-evidence dating analysis under the uniform tree prior and IGR model.

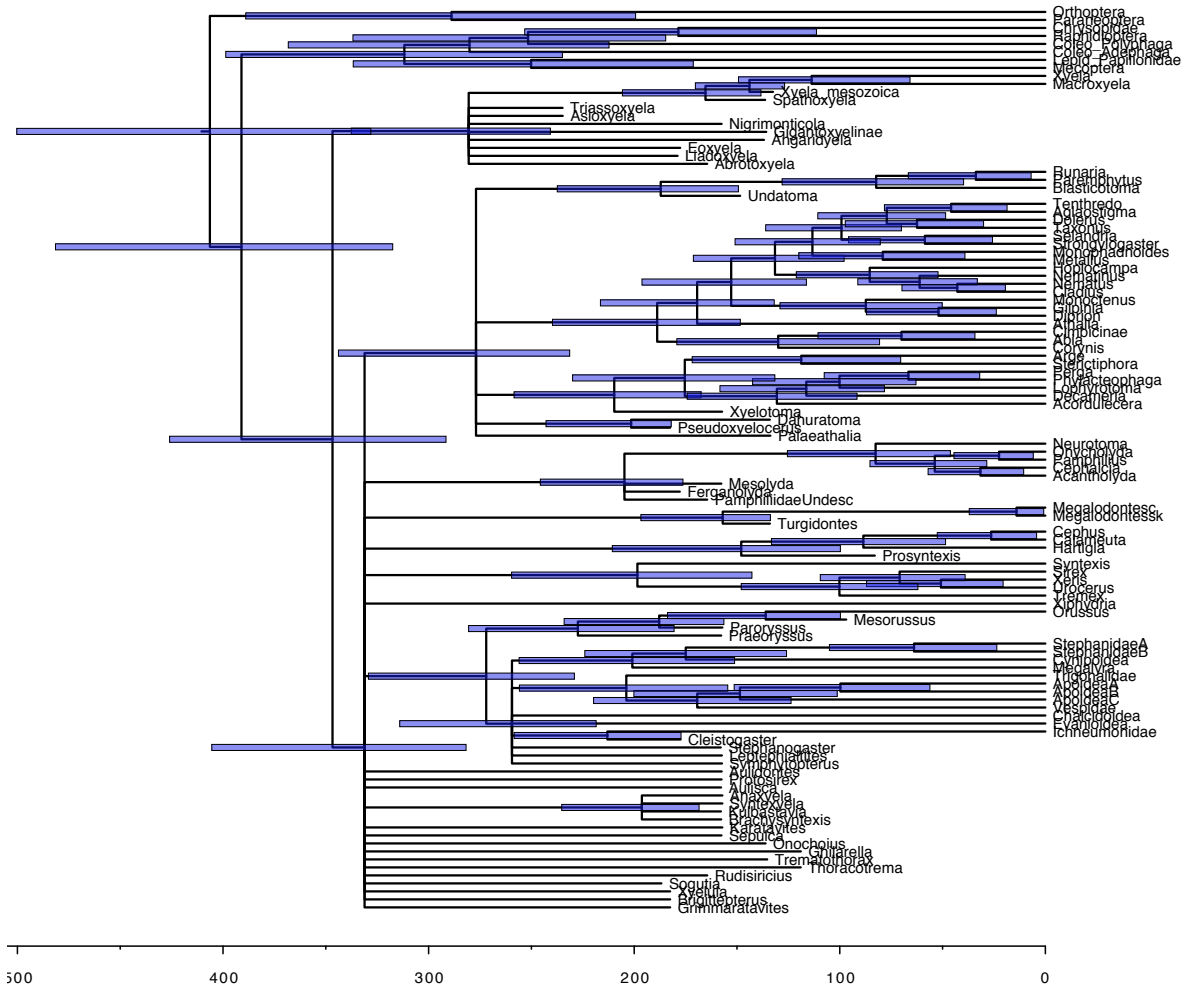


Figure S2: Majority-rule consensus tree from total-evidence dating analysis under the constant-rate random-sampling FBD prior and IGR model.

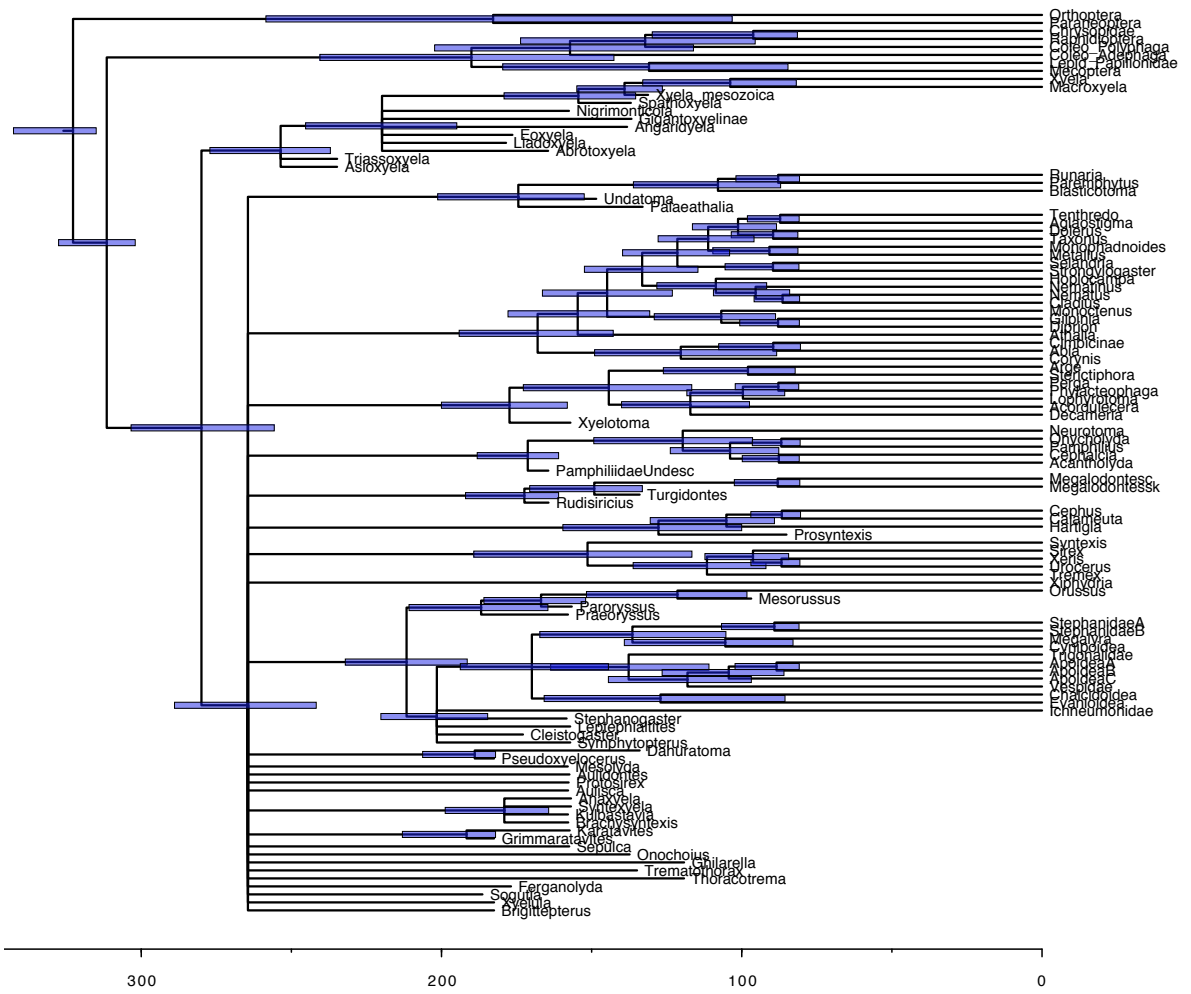


Figure S3: Majority-rule consensus tree from total-evidence dating analysis under the constant-rate diversified-sampling FBD prior and IGR model.

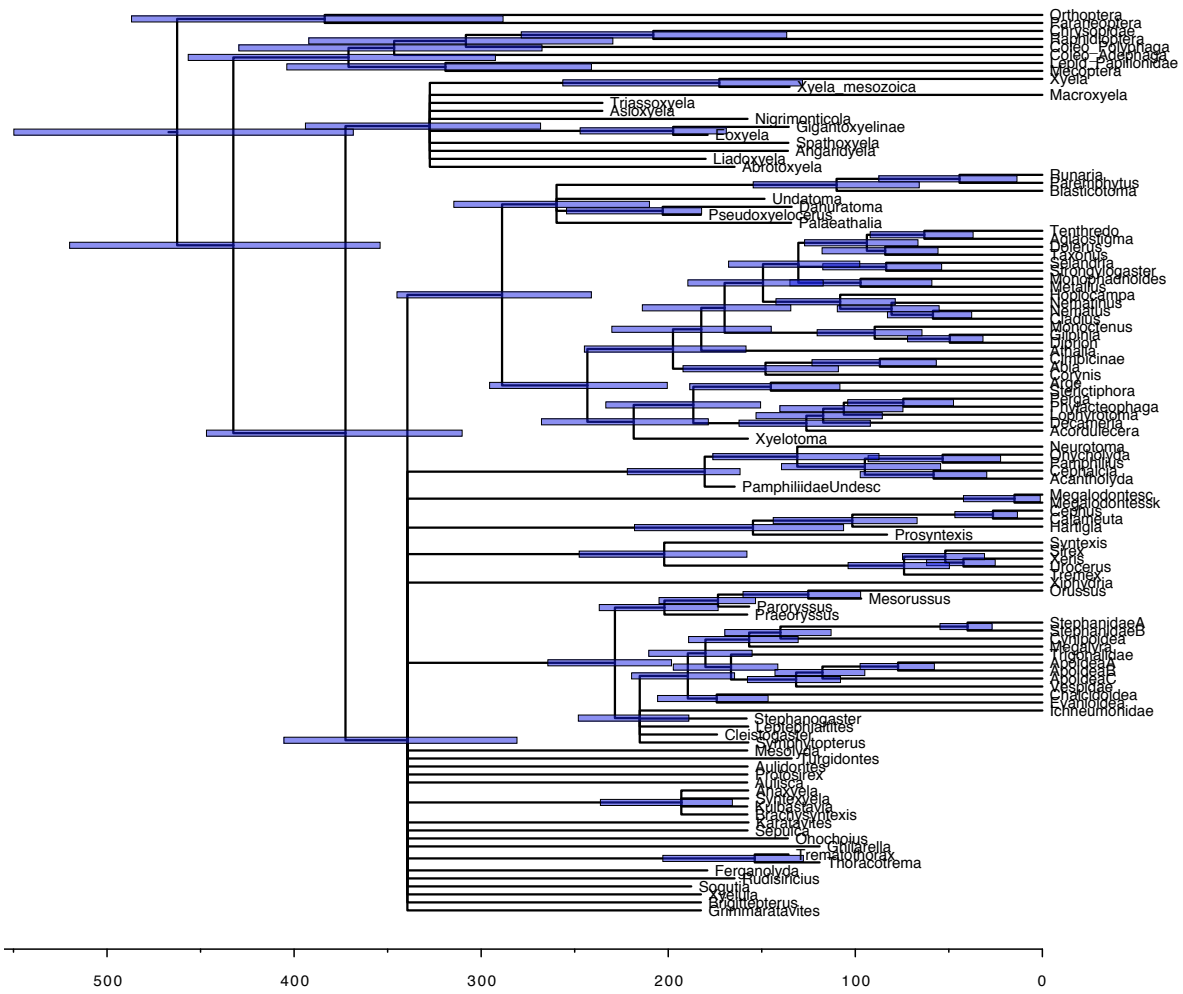


Figure S4: Majority-rule consensus tree from total-evidence dating analysis under the uniform tree prior and TK02 model.

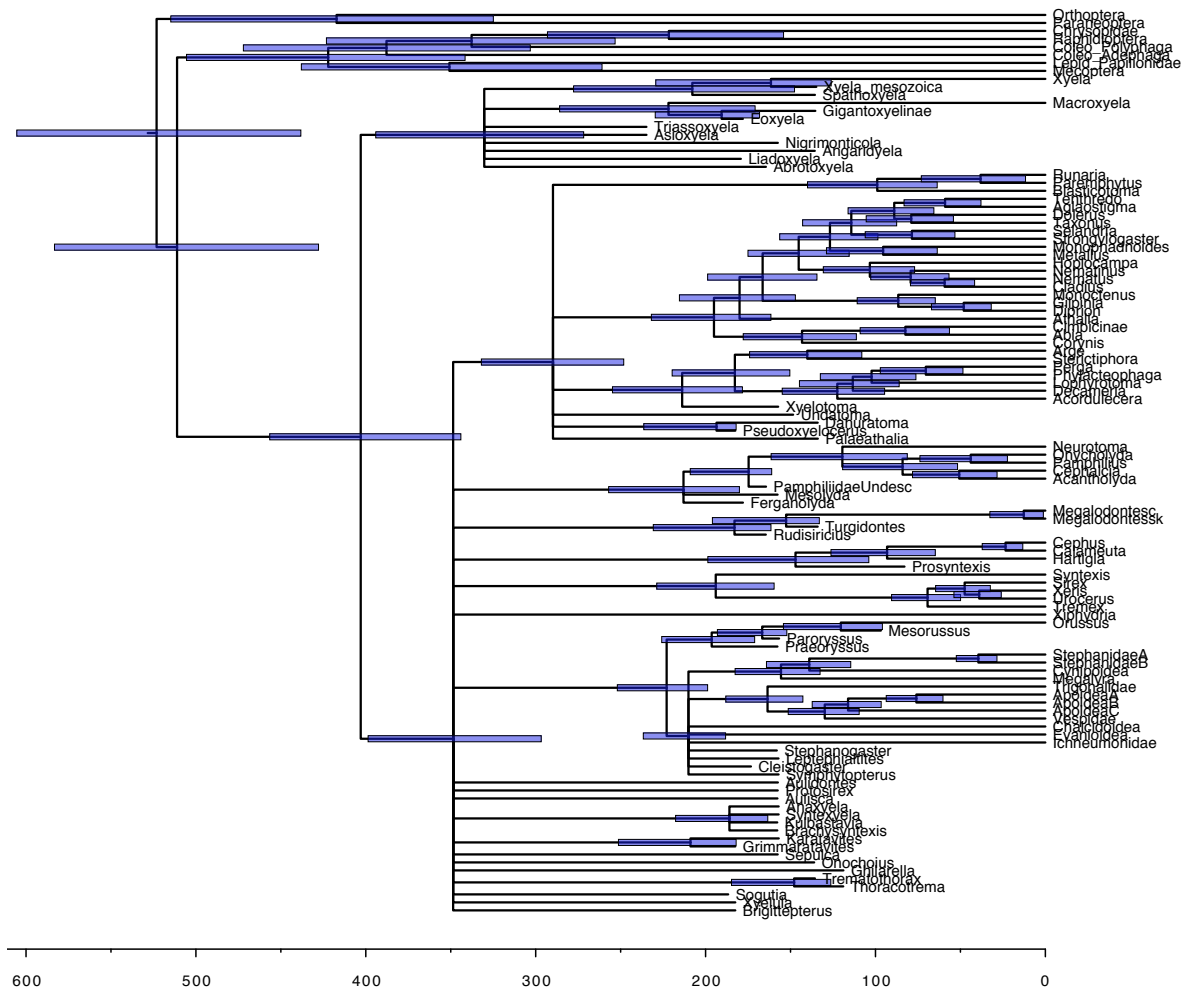


Figure S5: Majority-rule consensus tree from total-evidence dating analysis under the constant-rate random-sampling FBD prior and TK02 model.

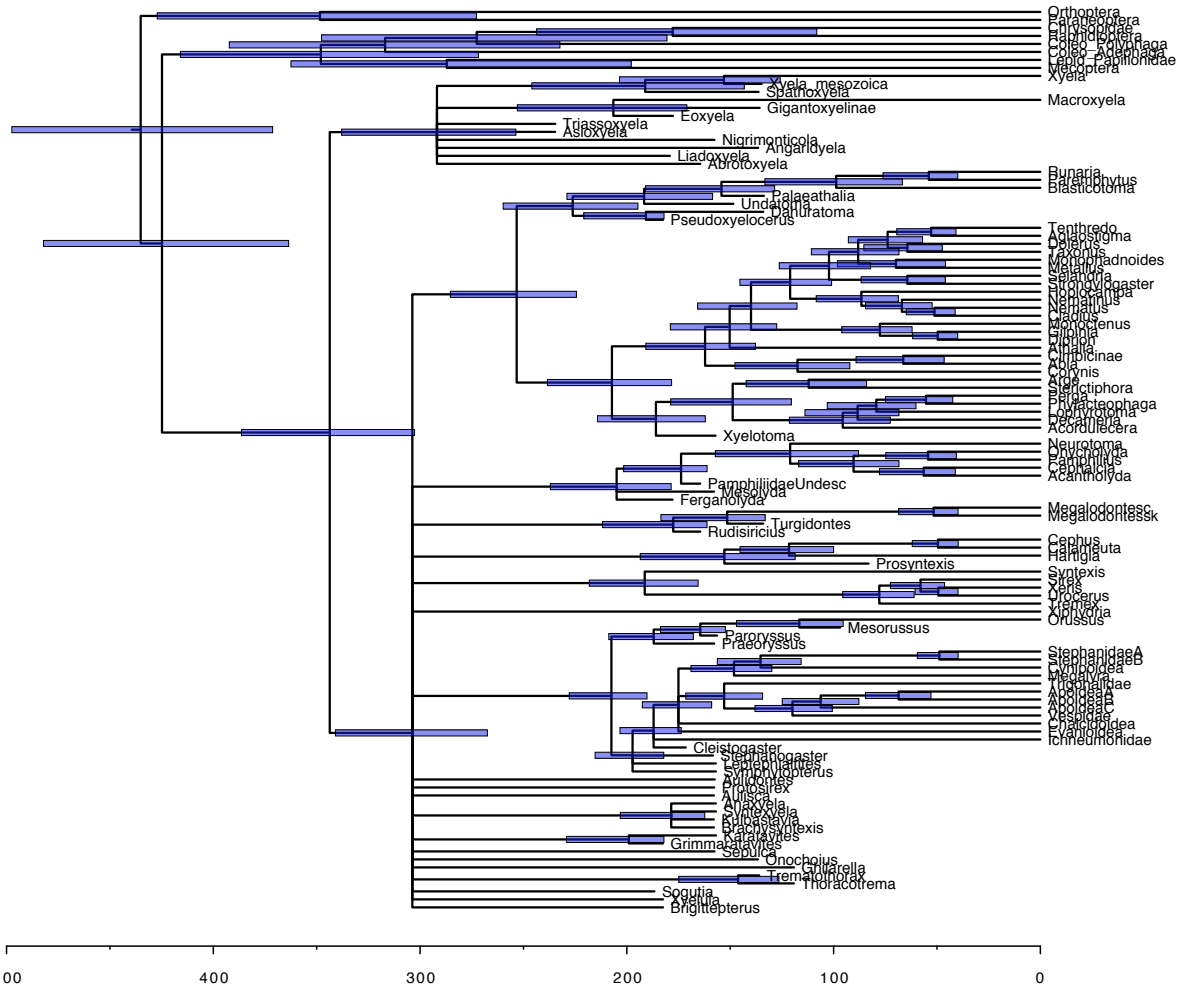


Figure S6: Majority-rule consensus tree from total-evidence dating analysis under the constant-rate diversified-sampling FBD prior and TK02 model.

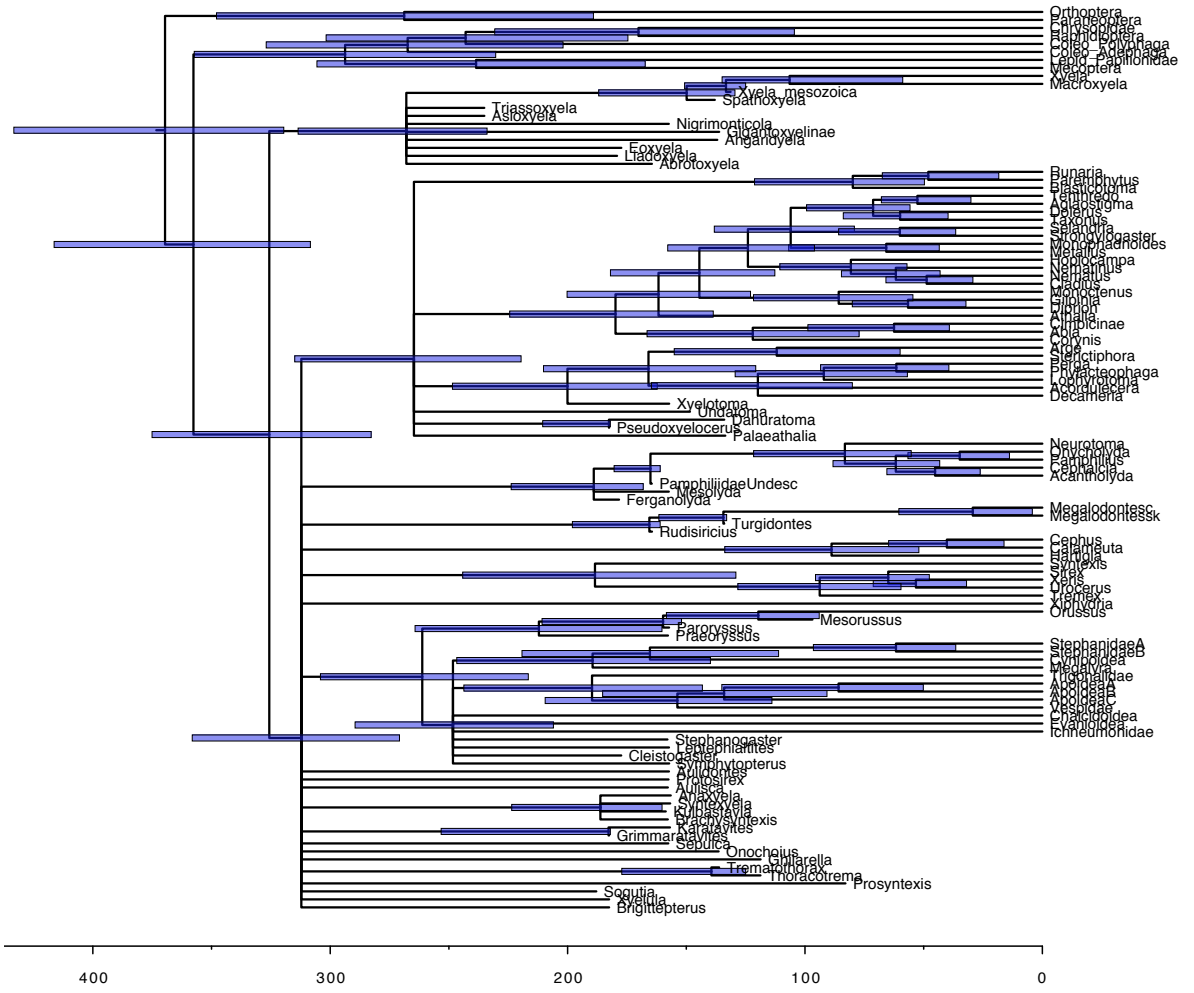


Figure S7: Majority-rule consensus tree from total-evidence dating analysis under the piecewise-constant random-sampling FBD prior and IGR model.

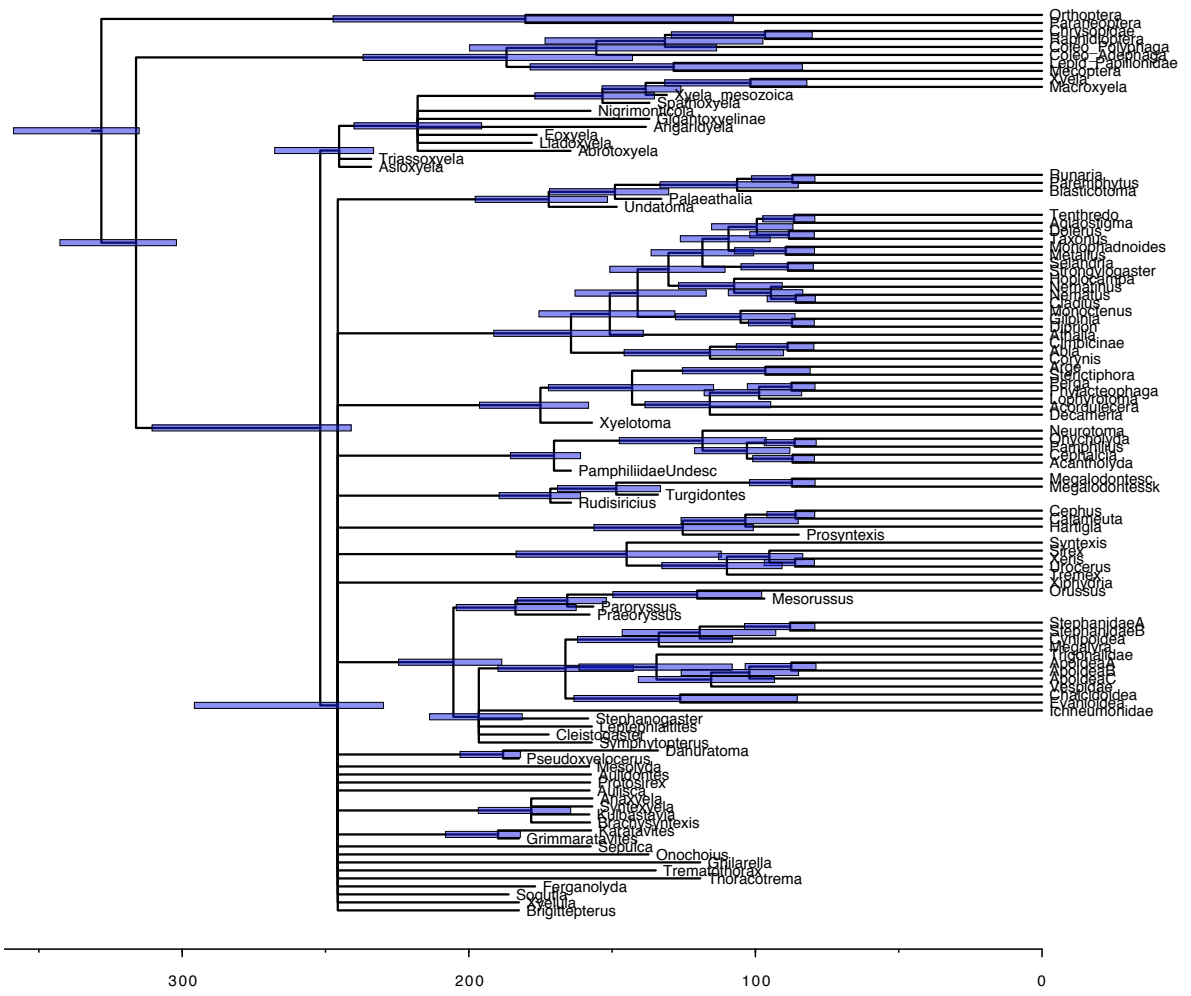


Figure S8: Majority-rule consensus tree from total-evidence dating analysis under the piecewise-constant diversified-sampling FBD prior and IGR model.

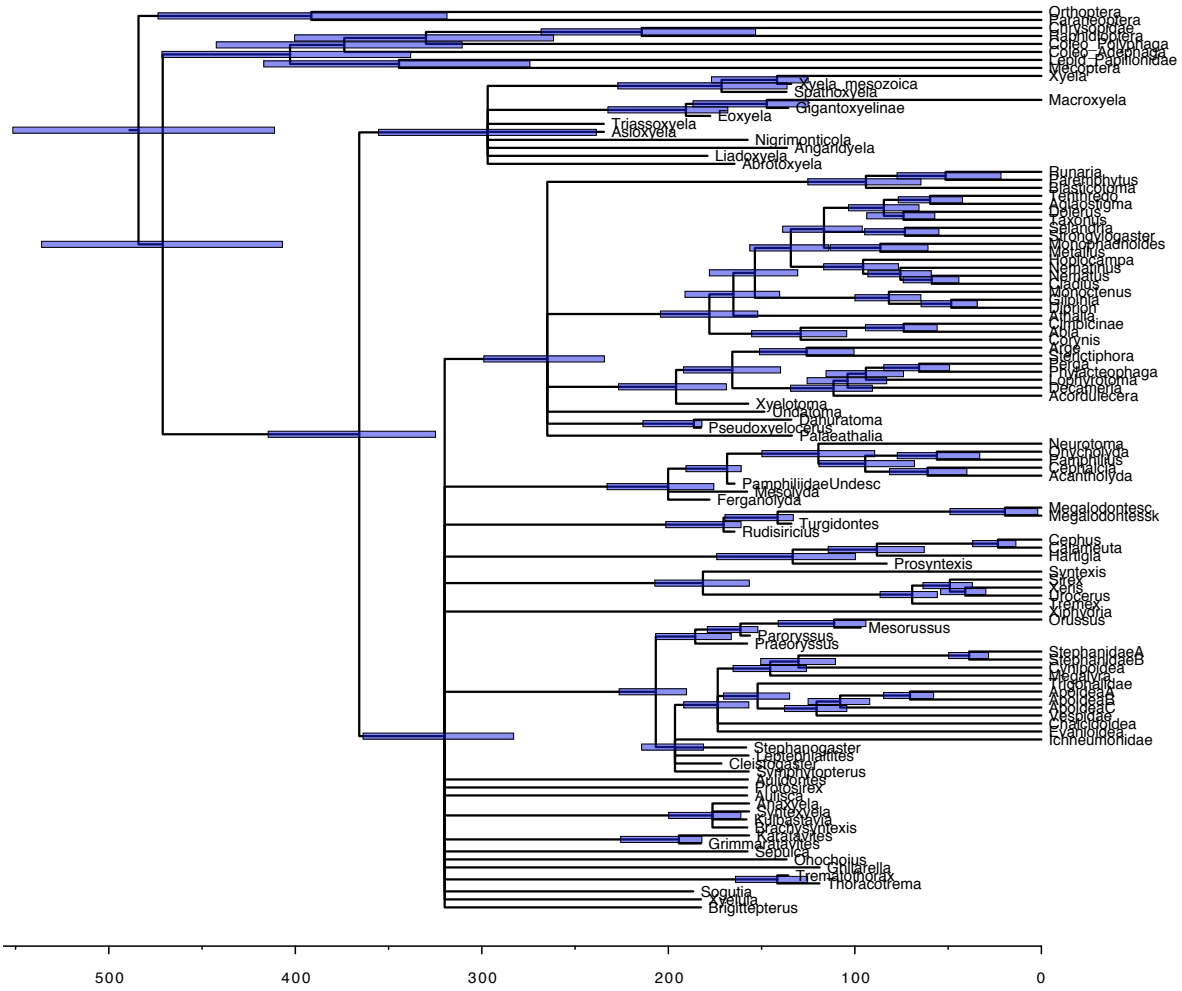


Figure S9: Majority-rule consensus tree from total-evidence dating analysis under the piecewise-constant random-sampling FBD prior and TK02 model.

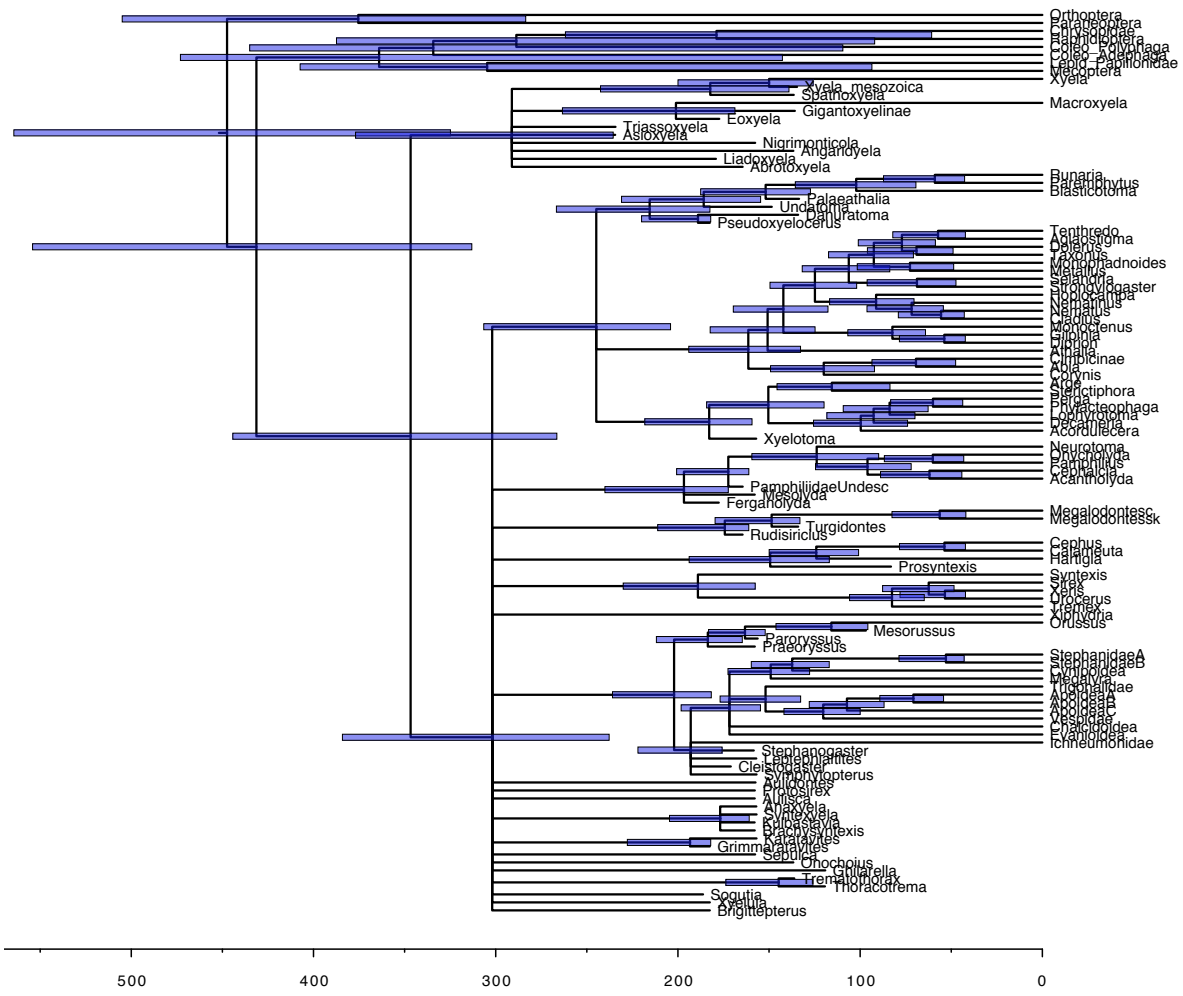


Figure S10: Majority-rule consensus tree from total-evidence dating analysis under the piecewise-constant diversified-sampling FBD prior and TK02 model.

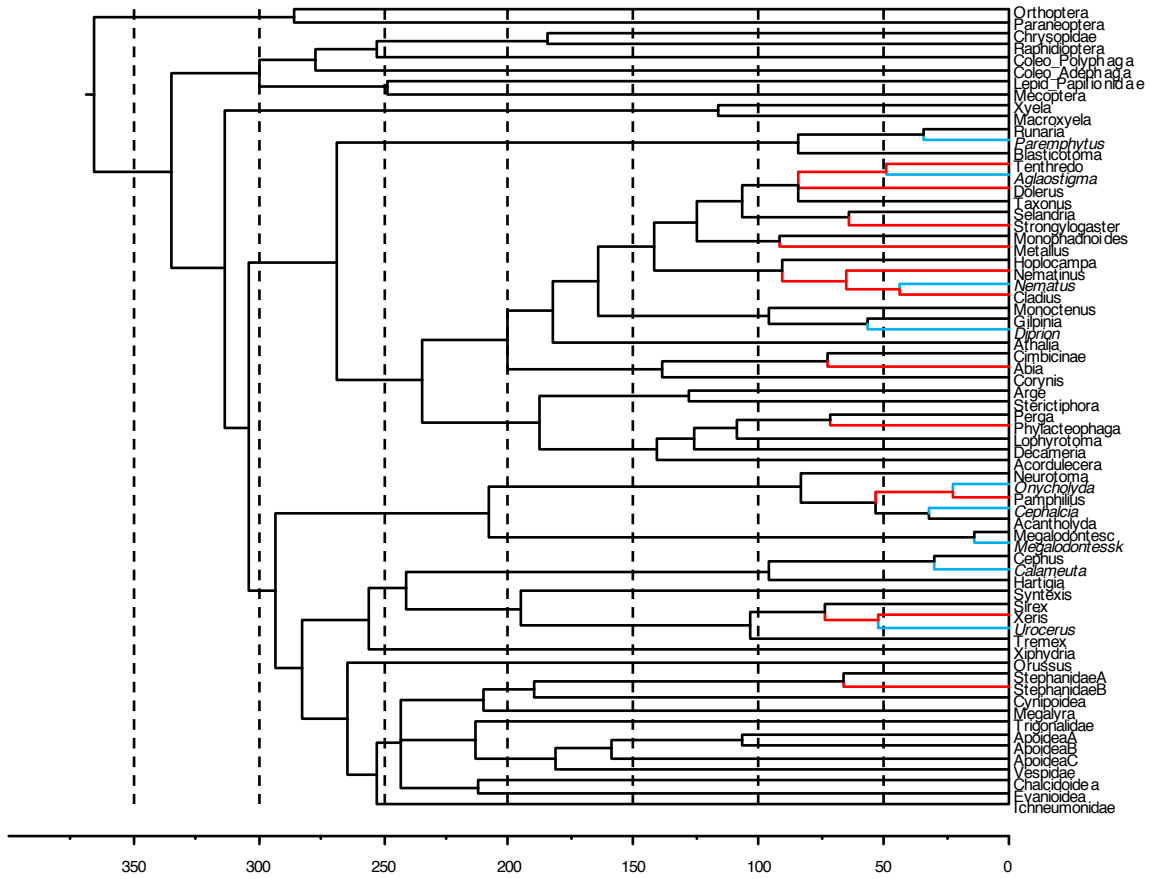


Figure S11: An uncalibrated relaxed-clock tree inferred under the uniform tree prior and IGR model. Two groups of lineages are removed from the tree, with 9 taxa (blue) and 20 taxa (blue and red) respectively. The rest of the taxa are used together with the fossils for additional analyses using the piecewise-constant diversified FBD model.